DP Computer Science: A 1.2.4 Truth Tables

1. Core Concepts & Learning Objectives

Learning Intentions

By the end of this lesson, students will be able to:

- **Construct** a truth table for a logic circuit of up to three inputs, with every input combination present in binary-count order
- **Construct** a truth table from a worded problem description by naming the inputs and output and translating the conditions into operators
- **Analyse** a truth table to read off its sum-of-products expression and to decide whether two circuits are logically equivalent
- **Simplify** an output expression using a **Karnaugh map** of up to three variables, forming groups that are powers of two

> **Command Term Note:** This lesson covers both **Construct** and **Analyse**. Karnaugh maps get more room than before—they are not easy for students and were rushed in the previous format. Boolean algebra laws and diagram construction move to A1.2.5. Four-variable maps are excluded (per the Computer Science FAQ document).

Key Vocabulary (Tier 3)

Term	Definition
Truth table	A table listing every input combination of a logic system and the output for each
Sum-of-products expression	An expression built by ORing together one AND term for each row whose output is 1
Logically equivalent	Two circuits are equivalent when their truth tables match for every input row
Karnaugh map	The truth table redrawn as a grid so adjacent cells differ in one variable, which lets 1s be grouped to simplify
Gray code	The axis labelling (00, 01, 11, 10) in which each label differs from its neighbour in one bit
Group	A block of adjacent 1s on the map whose size is a power of two, read as the variables that stay constant across it

Pedagogical Focus

- **Thinking Skills (ATL):** Instead of judging by eye whether two circuits behave the same, students build both truth tables and check every row, so the conclusion is settled by evidence. Name this to students as an **exhaustive check**.
- **Inquiry (ATT):** The teaching move here is to **withhold the answer**. Put the two circuits up, ask whether they are equivalent, and resist confirming it until the tables have decided. Explanation comes **after** the investigation, not before.

Name the skill to students: "*Today you are proving equivalence—not guessing it. The truth table is the tool that settles the question.*"

EAL Scaffold

Support Type	Description
Gate Symbol Reference	Seven gate symbols on display throughout

Support Type	Description
Binary-Count Skeleton	Input columns pre-filled (000, 001, 011, 010, 110, 111, 101, 100)
Karnaugh Grid	Axes pre-labelled in Gray code

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2. Lesson Sequence (One-Hour Block)

Part 1: Do Now (Retrieval) — ~10%

Activity: Two short recall items:

Question	Source
1. State the output of AND, OR, NOT, and XOR for each input pair	[A1.2.3]
2. How many rows does a complete truth table need for three inputs, and why?	[A1.2.1]

Part 2: Main Sequence — ~60%

Phase 1: Pose and Equip (~10 mins)

The Central Question:

> "*Circuit P and Circuit Q look different. Are they equivalent?*"

Equip Students With:

- Seven gate symbols on display as a reference
- A short worked example: constructing a truth table from a circuit, tracing gate by gate and labelling each intermediate signal

Inputs	Intermediate 1	Intermediate 2	Output
A	B	...	...

Phase 2: Investigate and Simplify (~20 mins)

Checkpoint 1 — Construct Truth Tables

Step	Task
1	Construct the full truth table for Circuit P using the binary-count skeleton
2	Construct the full truth table for Circuit Q
3	Compare the output columns **row for row**
4	Decide: **Are they equivalent?**
5	Read the **sum-of-products expression** from the table

Sum-of-Products Example:

A	B	C	Output
0	0	1	1
0	1	1	1
1	0	0	1

Expression: `F = A'B'C + A'BC + AB'C'`

Checkpoint 2 — Karnaugh Map Simplification

What is a Karnaugh Map?

- The truth table redrawn as a grid
- Adjacent cells differ in **one variable**
- Axis labels use **Gray code** (00, 01, 11, 10)

Two-Variable Worked Example:

	B=0	B=1
A=0	0	1
A=1	1	1

Grouping Rules:

- Groups must be **powers of two** (1, 2, 4, 8...)
- Groups should be as **large as possible**
- Groups can **wrap around** edges
- Groups can **overlap**

Three-Variable Layout:

	BC=00	BC=01	BC=11	BC=10
A=0				
A=1				

Reading a Group:

- Find the variables that stay **constant** across the group
- If constant = 1 → use the variable (e.g., A)
- If constant = 0 → use the complement (e.g., A')

Example:

- Group of two in bottom row (A=1, BC=01 and BC=11)
- A stays constant at 1
- B changes (0→1) → eliminated
- C stays constant at 1
- **Term:** `AC`

Phase 3: Explanation Consolidating (~5 mins)

Key Takeaways:

Principle	Explanation
Truth Table as Bridge	The truth table connects a circuit you can see to an expression you can manipulate
Equivalence	Two circuits are equivalent **exactly when** their tables match for every row
Karnaugh Map Groups	A group removes the variable that changes across it, which is why grouping simplifies

> "*The truth table is the exhaustive check—every row must match.*"

Part 3: Deliberate Practice — ~20%

Activity (Pairs):

Task	Description
Task 1	Construct a truth table from a short worded problem—name inputs and output, translate conditions into operators
Task 2	Simplify one three-variable expression with a Karnaugh map, naming each group

Worded Problem Example:

> "*A security system should trigger an alarm if the door is open AND the motion sensor is activated, OR if the window sensor is triggered.*"

Sentence Frame:

> "*The group at [A/B/C values] gives the term ______ because ______ changes and ______ stays constant.*"

Part 4: Exit Check — ~10%

Two Items (Readable in Minutes):

Question	Purpose
Part A: "Are these two short circuits equivalent? Give a one-line justification from the tables."	Checks whether students are proving equivalence or asserting it
Part B: "Simplify this three-variable expression with a Karnaugh map, naming the group."	Checks procedural application of simplification

> **Read Part A first** — it shows whether students are proving equivalence or asserting it.

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3. Karnaugh Map Reference

Two-Variable K-Map

	B=0	B=1
A=0	0	1
A=1	1	1

Simplified: `A + B`

Three-Variable K-Map

	BC=00	BC=01	BC=11	BC=10
A=0	0	0	1	0
A=1	1	1	1	0

Groups:

1. Horizontal group of two: `AC` (A=1, C=1, B changes)
2. Vertical group of two: `BC` (B=1, C=1, A changes)

Simplified: `F = AC + BC`

Grouping Rules Summary

Rule	Description
Group size	Must be a **power of two** (1, 2, 4, 8...)
Group shape	Rectangles or squares (not L-shapes)
Wrap-around	Groups can wrap around edges (top to bottom, left to right)
Overlap	Groups can overlap to make larger groups
Maximise	Use the **largest possible** groups
Variables eliminated	The variables that **change** within a group are eliminated
Variables kept	The variables that stay **constant** are kept

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4. Differentiation & Extension

Support (Removable Scaffolding)

Support Strategy	Description
Binary-Count Skeleton	Pre-drawn input columns (000, 001, 011, 010, 110, 111, 101, 100)
Gate Symbol Reference Strip	Seven gate symbols on a reference card
Karnaugh Grid	Axes pre-labelled in Gray code
Sentence Frame	For describing groups

> **Withdraw scaffolding** once students can set them down themselves.

Challenge (Deeper and Wider, Not Merely More)

Challenge Strategy	Description
Spot the Mismatch	Give a third circuit that looks equivalent but differs in exactly one row—students find the mismatch and state where the behaviour parts
Group Verification	Confirm that a group read off the map matches what factoring the expression by hand would give
Wrap-Around & Overlap	Use wrap-around and overlap on a three-variable map to form the largest legal group

> **Extension stays within three variables and the Analyse band.**

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5. Example Hinge Questions

Question	Purpose
"How many rows does a complete truth table need for three inputs, and why?"	Checks understanding of binary-count principle
"What does it mean for two circuits to be logically equivalent?"	Checks understanding of equivalence definition
"Why do we use Gray code for Karnaugh map axes?"	Checks understanding of adjacent-cell principle
"When reading a group on a Karnaugh map, which variables are eliminated and why?"	Checks understanding of grouping logic
"What is the key difference between an OR gate and an XOR gate?"	Checks understanding of confusable pair

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6. Learning Objectives (Summary)

Knowledge and Understanding

- Construct truth tables for logic circuits with up to three inputs
- Analyse truth tables to determine output for any input combination
- Read a sum-of-products expression from a truth table
- Use Karnaugh maps to simplify Boolean expressions
- Determine logical equivalence using truth tables

ATL Skills

Skill	Description
Thinking	Apply logical reasoning to prove equivalence; identify patterns for grouping
Communication	Explain simplification steps using precise terminology
Self-Management	Organise and complete multi-stage tasks

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7. Resources

- **Presentation** (Slide Deck)
- **InThinking eBook** (A1.2.4)
- **Worksheet:** Truth table construction + Karnaugh map simplification
- **Worksheet Answers**

Author's Reflection

> *"The main change here is the format. The material I started from ran the content as a set of timed team games—a relay race, a triathlon, and a Karnaugh kart race. The tasks underneath were sound, so I kept them and dropped the competition, which rewarded speed rather than the reasoning the content needs. In its place the lesson is one question: are these two circuits equivalent, and if so why? Students build the truth table for each, compare them, and then simplify to show why, which gives the tables and simplification a clear purpose.*

> *Karnaugh maps get more room than before. They are not easy for students and were rushed in the race format, so here they are taught with their own worked examples and practice, and the Boolean-algebra laws and diagram construction move to the next lesson (A1.2.5) so simplification is not taught twice. I also took out the four-variable maps that the original answer key used, since the scope note excludes them (Computer Science FAQ document), so every map in this lesson stays at three variables."*